

ANALYSIS OF THE OPTICAL FLOW OF THE EYE USING FUNCTIONAL MAGNETIC RESONANCE IMAGING

Chathura Jayasekara
Degree Programme of Biomedical Engineering
University of Oulu

2025

Contents

1	Introduction	2
2	Background	5
2.1	Invasive studies	5
2.2	Non-invasive studies	7
2.3	Current ocular glymphatic models	10
3	Objective of the study	12
4	Materials and Methods	13
4.1	Data Aquisition	13
4.2	Preprocessing	14
4.3	Optical flow analysis	14
5	Results	16
5.1	Analysis on VLF band	16
5.2	Analysis on RESP band	19
5.3	Comparison of VLF and RESP bands in retina and optic nerve	23
6	Discussion	27
7	Conclusions	29

1 Introduction

The human eye generates substantial waste, including metabolic waste and photoreceptor by-products, particularly near the retina. Despite this, it lacks the conventional lymphatic vessel system [1], which is responsible for effective waste removal and maintaining fluid homeostasis in most other tissues. Therefore, the eye should rely on an alternative waste clearance mechanism. A similar situation exists with the human brain, which was once thought to be devoid of such a waste-clearance system, despite its exceptionally high metabolic activity. For decades, many researchers have suggested the existence of lymphatic-like vessel systems in the brain, and in 2012, for the first time, Iliff et al. conceptualised the "cerebral glymphatic clearance system" [2]. Subsequent studies provided evidence supporting the existence of such an effective removal system within the brain. In 2020, Wang et al. proposed a similar alternative waste clearance mechanism for human eye, as "ocular glymphatic clearance system" [3].

The brain glymphatic system differs anatomically from the conventional lymphatic vessel system of the rest of the body, as it uses existing spaces in the brain, such as the perivascular paths of cerebrospinal fluid (CSF) [1], [4]. The CSF is conveyed from the peri-arterial spaces to the interstitial space via the aquaporin-4 (AQP4) water channels of the astrocytic endfeet. The interstitial space is filled with interstitial fluid (ISF), which collects metabolic by-products generated by neurons and glial cells. Research indicates that ISF may exchange solutes and waste materials with the CSF. Finally, the CSF-ISF mixture, along with the collected waste, is subsequently cleared from the interstitial space into the perivenous spaces and further mediated to the meningeal lymphatic vessels and cervical lymph nodes. The question has been the driving mechanism of this proposed glymphatic clearance system of the brain. Experimental evidence [5], [6] indicates that glymphatic flow is driven mainly by physiological pulsations, including respiratory (RESP), cardiac (CARD), and Very Low Frequency (VLF) pulsations associated with vascular dynamics. This model proposes a mechanistic methodology for the existence of an effective waste clearance and fluid homeostasis pathway in the brain, despite its absence of a conventional lymphatic vessel system.

Recent clinical evidences support the existence of an ocular glymphatic system [3], [7]. Further, Wang et al. in [7] suggested that ocular waste clearance may be integrated with the cerebral glymphatic system, as the optic nerve can be considered an extension of the central nervous system. Several physiological observations further support this concept. The retina exhibits considerable metabolic energy consumption [8], even during the eye-closure state, requires substantial oxygen and glucose for photopigment regeneration, and shows significant metabolic activity in the retinal pigment epithelium and Muller glia. In

addition, the retina has a dual blood supply, reflecting a high metabolic demand. Further, variation in intraocular pressure (IOP) are highly associated with ocular pathologies such as such as glaucoma, ocular hypertension [3] , [1]. Traditionally, the regulation of IOP and maintenance of fluid homeostasis were believed to be achieved through the trabecular and uveoscleral aqueous humour pathways. However, this mechanism does not account for ocular waste clearance, such as amyloid and tau proteins. These observations suggest that there must be an efficient metabolic waste clearance mechanism within the eye. Nonetheless, this remains a hypothesis that requires further investigation.

Study of fluid movement is crucial for understanding the ocular glymphatic system functionality. There are several invasive approaches, such as animal tracer studies (mostly mice) and human postmortem studies, which are essential for establishing the anatomical plausibility of the fluid pathways. However, this lacks the insight into the glymphatic system fluid dynamics and the driving mechanisms. Further it is limited to human in vivo studies since tracers can disturb the microvasculature of the eye and optic nerve. Mainly due to this gap, non-invasive approaches were developed, ranging from functional Magnetic Resonance Imaging (fMRI) based methods to ophthalmic imaging and tear biomarker studies, which allow human in vivo testing. Building on these non-invasive approaches, our study is based on functional eye MRI imaging (fMREye) data captured from healthy subjects. Then, we applied optical flow analysis based on the Lucas-Kanade (LK) approach on fMREye data to uniquely captures the dynamic spatiotemporal propagation of ocular and cerebral fluid pulsations. In this way, we aimed to provide a novel functional readout of ocular glymphatic activity.

The primary aim of this study was to investigate the driving mechanisms and pulsatility of the human ocular glymphatic system by analysing optical flow in the human eye using fMREye data previously collected by the Oulu Functional Neuroimaging (OFNI) group. All analyses were conducted within the OFNI group facility. This approach enabled the characterisation of ocular pulsatility, which may represent a driving force for fluid transport through the previously proposed clearance pathways. Ultimately, the results may provide insights into the ocular waste clearance system, as suggested by previous studies. During this study, two well-established pulsation bands, VLF and RESP, were examined, following the methodology of earlier work from the OFNI group [6], [9]. Elucidating the mechanisms and driving forces of ocular metabolic waste clearance is crucial for advancing both physiological and pathophysiological understanding.

This study was conducted within the OFNI group using its facilities, including licensed software platforms such as MATLAB and FSLeys. The data were collected and pre-processed by the group and were thus ready for analysis. The author developed and applied custom MATLAB code for data analysis, comparison, and visualisation of the results. Artificial intelligence tools were employed for spelling verification, while the writing was carried out by the author with reference to the available literature, where

appropriate (cited accordingly).

2 Background

Although the waste clearance system of the human eye has been a topic of discussion for several years, most research studies have been limited to tracer-based rodent studies and post-mortem human eye studies, both of which are invasive. Tracer-based invasive methods are not feasible for in vivo human eye studies, as the tracers may obstruct the nanovasculature of the eye, potentially leading to complications, including blindness. Consequently, available tracer studies have been performed on rodents [3], [7], [4]. These studies have revealed potential fluid pathways and based on results murine models of ocular waste clearance have been proposed.

In addition, postmortem studies have been performed on human eyes as well as the brain [4], where lymphatic-like vasculatures were observed in the eye, particularly in the ciliary body. However, these postmortem studies were primarily based on immunohistochemical approaches where biomarkers of lymphatic vessels were identified. However, similar biomarker expression may occur in non-lymphatic tissues, which affects the interpretation of the results.

Recently, increasing focus has been given towards noninvasive studies such as fMRI in order to address the limitations of invasive studies. Noninvasive methods are fast, reproducible on the same subject and can be performed under various physiological conditions, enabling investigation of the behaviour of the waste removal process under more natural circumstances. This has led to a plethora of research opportunities, including how pupil constriction affects waste removal and how sleep affects ocular waste clearance. Importantly, unlike invasive methods, these methods do not interfere with the eye vasculature functionality, thus preserving normal physiological behaviour. As a result, experiments can be conducted on healthy subjects without any side effects.

For the clarity of comparison and understanding, the available literature is discussed in depth separately as invasive studies and non-invasive modern studies.

2.1 Invasive studies

In 2012, Nedergaard et al. proposed the glymphatic system as the brain's waste clearance pathway [2]. In 2015, Denniston and Keane hypothesised an analogous mechanism in the eye, termed the ocular glymphatic system [10]. However, Wang et al. provided experimental evidence for such a pathway in rodents using a tracer study in 2020[3]. They injected HiLyte-594-tagged human β -amyloid (hA β) into the vitreous body of mice and, after one hour, detected its entry into the optic nerve sheath through the glial lamina at the head of the optic nerve, indicating an alternative route to the conventional

anterior pathway. Subsequently, they injected a second tracer into the cisterna magna, a component of the brain's CSF pathway, and observed retrograde transport through the optic nerve sheath. The largely unmixed distribution of the two tracers suggested distinct routes for anterograde and retrograde fluid movement [3, 11]. This ocular clearance route is analogous to the brain's glymphatic system and, as demonstrated in the same study, depends on AQP4 water channels in glial cells. In 2023, the same group [11] conducted a tracer study in mice to investigate the effects of ageing on the ocular glymphatic system. This study confirmed their previous observation of retinal waste clearance through the optic nerve with bidirectional flow. They also reported that natural ageing impairs ocular glymphatic clearance, similar to the decline observed in the brain's glymphatic system. Furthermore, they found that under glaucomatous conditions, both the anterograde and retrograde ocular pathways are considerably altered, resulting in increased fluid flow. So, ageing and glaucoma affect ocular glymphatic clearance in two distinct ways. These observations may prove crucial for the identification of pathological conditions.

Recent human postmortem studies have yielded several notable observations. H. E. Killer et al. [12] performed a postmortem study on the optic nerve sheath dura mater of human cadavers to investigate the presence of lymphatic vessels. Their investigations, based on electron microscopy and immunohistochemistry, reported lymphatic capillaries frequently in the bulbar segment of the optic nerve but only rarely in the subarachnoid space. They also claimed to observe lymphatic capillaries in the dura using transmission electron microscopy. However, this study did not identify any lymphatic vessels in intraocular structures such as the retina or vitreous. These regions were long considered avascular and alymphatic, which later contributed to the proposal of the glymphatic clearance system. In an immunohistochemical study, Yücel et al. observed lymphatic markers [13], suggesting the presence of lymphatic-like vessels in the ciliary body stroma. More recently, in 2024, Xiangyun et al.[14] conducted a tracer study in mice to investigate the drainage of the anterior and posterior compartments of the eye. Their findings suggested a compartmentalised drainage system in which aqueous humour from the posterior eye drains to the superficial cervical lymph nodes (sCLNs), whereas vitreous humour from the posterior eye drains to the deep cervical lymph nodes (dCLNs). These results led to the conclusion that the two chambers have independent drainage routes and that interference with one pathway does not significantly affect the other. Furthermore, they employed a transcriptomic approach, including RNA analysis, to identify genes indicating lymphatic-like properties, combined with immunolabeling-enabled 3D imaging of solvent-cleared organs (iDISCO). Their results confirmed the presence of lymphatic-like vasculature in the trabecular meshwork and retina, as well as lymphatic- or glymphatic-like vessels in the optic nerve sheath. However, these findings do not confirm the existence of traditional lymphatic vessels but instead suggest the presence of lymphatic-like pathways that might facilitate waste and fluid clearance from the eye through the optic nerve sheath.

In summary, researchers have proposed several waste removal pathways using various experimental methods such as tracer studies, immunohistochemical approaches, and RNA analyses. However, tracer studies conducted in mice have certain anatomical differences compared to the human eye. For instance, rodents have a glial lamina, whereas humans possess a lamina cribrosa, which may exhibit distinct physiological properties. RNA and immunohistochemical methods rely on biomarkers relevant to specific anatomical features, but similar positive observations can be produced by different biomarkers, making it difficult to confirm the exact structures involved in the observation. Moreover, tracer studies can interfere with normal physiological processes due to the nanovascularity, and the optic nerve must often be harvested after tracer administration to identify tracer spread [9]. Postmortem studies are also limited, as physiological processes may be altered after death. Importantly, both tracer- and postmortem-based methods do not allow for true *in vivo* testing, particularly in healthy subjects, making it difficult to compare normal and pathological conditions. Some of these shortcomings can be addressed through noninvasive imaging methods, which provide dynamic information rather than simply confirming the anatomical structure of clearance pathways.

2.2 Non-invasive studies

Several non-invasive methods have been proposed for studying the human ocular lymphatic system, including MRI-based techniques, EEG/MEG-combined approaches, and ophthalmic OCT. PET imaging with tracers can also be used, but it is considered minimally invasive. Among these, MRI-based techniques have become essential for both ocular and cerebral lymphatic research.

MRI employs powerful magnetic fields together with radio-frequency pulses to produce detailed anatomical scans, avoiding the ionising radiation required in PET and CT. Varying T1, T2, or proton-density settings allows MRI to emphasise different tissue properties while maintaining very fine spatial detail (0.1–1.0 mm³). The technique’s main limitation is its slow sampling rate: conventional anatomical MRI takes too long to acquire images to follow rapid physiological events. Consequently, high-frequency phenomena, like CARD (1 Hz) and RESP (0.2–0.4 Hz) cycles, are aliased, therefore reducing the reliability of measurement of dynamic processes associated with cerebral or ocular lymphatic clearance.

Functional MRI (fMRI) is designed to detect dynamic neural and physiological changes using the blood-oxygen-level-dependent (BOLD) signal, which captures fluctuations in hemoglobin oxygenation driven by local neuronal processes. Typical fMRI acquisitions achieve temporal resolutions of 1–3 seconds per whole brain, which is faster than anatomical MRI but still not capable of resolving rapid physiological rhythms without aliasing. Although fMRI provides valuable information about neural networks, resting-state ac-

tivity, and hemodynamics, its temporal resolution remains insufficient for studying the high-frequency drivers of glymphatic transport. To study these dynamics, much faster acquisition sequences are required. Echo-Planar Imaging (EPI) is a rapid/fast fMRI technique capable of collecting an entire 2D slice in a few shots and is the standard sequence for BOLD-fMRI. EPI can acquire whole-brain volumes in approximately 1–3 seconds, enabling improved temporal tracking of neural processes. However, this speed is still inadequate for accurately resolving CARD and RESP oscillations.

Magnetic Resonance Encephalography (MREG) is an ultra-fast 3D undersampled acquisition technique that overcomes these limitations. MREG can acquire whole-brain BOLD volumes with a temporal resolution of 10–100 ms (sampling rates up to 10 Hz) with 3D single-shot spiral trajectory. This enables reliable measurement of physiological pulsations across several frequency bands, including CARD, RESP, and VLF oscillations. The high sampling rate dramatically reduces cardio-respiratory aliasing and allows detection of subtle tissue displacement and brain-wide pulsation dynamics that may contribute to ocular and cerebral glymphatic transport. A limitation of MREG is its lower spatial resolution (3–5 mm isotropic), which provides reduced anatomical detail. Therefore, MREG data are commonly complemented with high-resolution T2-weighted anatomical images, with the two modalities aligned using standard image-registration pipelines.

Although MREG is well suited for assessing global cerebral dynamics, its 3D spiral readout is not ideal for resolving fine anatomical features within the eye. To address this, specialised rapid fMRI method such as fMREye can be used. fMREye can capture fast physiological fluctuations in ocular tissues [9] by using EPI over one or a few neighbouring slices. These protocols acquire images with high in-plane resolution, allowing small intra-ocular structures to be clearly distinguished. At the same time, they provide sufficiently high temporal sampling (around 100 ms) to record physiological pulsations without aliasing. A key limitation of the current fMREye implementation is its very limited slice coverage. Expanding the number of slices or increasing slice thickness would improve volumetric coverage but would reduce spatial resolution and increase partial-volume effects, potentially lowering measurement accuracy.

A number of analytical approaches have been developed to study glymphatic flow and its driving pulsations using MRI data. Examples include perivascular diffusion analysis using the DTI-ALPS method and phase-contrast MRI [5], [4]. Optical flow analysis has recently emerged as a novel computational method for assessing motion dynamics in glymphatic research. Originally developed in computer vision for motion and velocity estimation, optical flow has not yet been reported for studying the ocular glymphatic system. However, in 2019, Laura et al. [6] from the OFNI group applied optical flow (Lucas–Kanade) to MREG data and successfully mapped three-dimensional cardiac pulse propagation throughout the brain. Their results closely matched known anatomical structures, demonstrating the feasibility of noninvasively mapping pulsatile brain motion. This

finding suggests that optical flow could similarly be extended to investigate pulse propagation in the eye and optic nerve, potentially enabling noninvasive assessment of ocular glymphatic pathways.

Optical flow analysis estimates motion by identifying changes in intensity patterns between consecutive images. In neuroimaging, this can provide a powerful means to quantify small dynamic changes in tissue or fluid that are not visible in static MRI. Optical flow computes voxel-wise motion between successive frames by applying algorithm to the scan data, resulting in a 3D velocity vector for each voxel. The resulting 5D optical flow dataset consists of voxel position (x, y, z), velocity direction (dx, dy, dz), and time. From this vector field, metrics such as flow magnitude, directionality, speed maps, and temporal correlations can be derived. With region-of-interest (ROI) definitions, ROI-averaged motion time series can be extracted for group-level statistical analysis. Thus, optical flow algorithm transforms raw scan data into a quantitative representation of ocular–cerebral pulsation and fluid motion, enabling physiologically meaningful interpretation and integration into mathematical or biophysical models.

Beyond MREG-based motion analysis, additional multimodal approaches continue to advance ocular glymphatic research. In 2024, Ebrahimi et al.[9] conducted a multimodal non-invasive imaging study using functional eye MRI (fMREye), an eye-tracking functional eye camera (FEC), and MRI-compatible video ophthalmoscopy (MRcVO). They identified CARD, RESP, and vasomotor (VLF) pulsatility components on the eye surface and in the retina, demonstrating that the eye exhibits not only CARD pulsations—as previously believed—but also strong RESP and VLF components. More recently, Van der Thiel et al. [15] proposed a multimodal framework combining MRI structural imaging, tear-fluid biomarkers, and intraocular pressure monitoring to study human glymphatic function. Their findings suggest a link between ocular and cerebral glymphatic clearance. For example, elevated tear tau concentrations may reflect impaired cerebrospinal fluid (CSF) clearance. This integration of tear biomarkers with MRI provides a promising noninvasive approach to studying glymphatic mechanisms and related pathologies. Furthermore, Levendovszky et al. [16] used intravoxel incoherent motion (IVIM) diffusion MRI to investigate neurofluid dynamics—a technique capable of assessing microvascular perfusion and tissue water mobility. Because glymphatic transport involves microscale fluid flow and ISF–CSF exchange, IVIM offers particular relevance. Their results showed that sleep deprivation reduces pseudodiffusion, a parameter believed to reflect perivascular pulsatility, one of the major driving forces of glymphatic clearance. These findings align with animal studies and suggest that IVIM-based methods may be extended to examine ocular glymphatic function.

2.3 Current ocular glymphatic models

Ocular glymphatic clearance models have been developed with collaborative contributions from both invasive and non-invasive studies. The earlier fluid clearance models, the conventional trabecular pathway and the uveoscleral pathway, do not adequately explain how high metabolic waste clearance occurs from the interior of the eye [1], although they do explain pressure regulation and fluid homeostasis in the anterior eye. The bidirectional clearance pathway model is one of the most promising ocular glymphatic models, as discussed in [11]. In this model, bidirectional flow occurs across the optic nerve sheath: anterogradely, cerebrospinal fluid (CSF) flows from the subarachnoid space through the peri-arterial space toward the retina, while retrogradely, retinal waste such as Beta -amyloid travels through the perivenous space. The transport is believed to occur across the glial lamina, with mixing and transfer facilitated by aquaporin-4 (AQP4) water channels. Studies [2] have shown that mislocalization and dysfunction of AQP4 (e.g., in knockout mice) result in reduced efficiency of this transfer. Most importantly, this dysfunction has been related to ocular diseases such as glaucoma and retinal degeneration, and possibly to brain diseases such as Alzheimer’s disease and age-related cognitive decline that may be connected with impaired ocular waste clearance. The driving mechanisms of this bidirectional model are thought to include arterial pulsatility and the translaminal pressure gradient, which is the difference between intraocular pressure (IOP) and intracranial pressure (ICP). This has been shown mainly in rodent studies, although pulsatility has also been supported by MRI studies. Further, studies suggested that retinal activities such as pupil constriction and dilation also influence intraocular fluid dynamics [9].

Another suggested model, mainly discussed in [1], is the lymphatic drainage pathway. However, this is essentially an extension of the above-mentioned glymphatic model. It is generally believed that ocular waste clearance lacks conventional lymphatic vessel systems. Recent studies [12], [17] have suggested that lymphatic-like vessels are visible in certain regions such as extraocular fat tissues, muscles, and the lacrimal gland, which may serve as potential ancillary drainage routes. In addition, lymphatic vessels have been reported in the optic nerve dura mater, suggesting that durally located lymphatics could contribute to posterior ocular clearance. Nevertheless, it is still not confirmed whether they function in the same way as conventional lymphatic vessels. In particular, the driving mechanisms through these lymphatic-like vessels remain unclear [12]. Therefore, it is suggested that these vessels functionally support the proposed ocular glymphatic system.

From these studies and models, it is becoming clearer that the ocular waste clearance system and the cerebral waste clearance system are functionally interconnected, meaning that insights from one system can support the study of the other. Further, dysfunction of the ocular clearance system certainly contributes to ocular diseases such as glaucoma,

ocular hypertension, retinal degeneration, and age-related macular degeneration. Furthermore, studies have revealed that in brain diseases such as Alzheimer's disease and Parkinson's disease, the ocular glymphatic clearance system shows considerable alterations [3], [1]. However, based on the current literature, it is not clear whether these observed defects contribute to the development of the disease or whether they are consequences of it. What is clear is that variations in the ocular glymphatic system can serve as a window into the identification of brain diseases [1]. For instance, in Alzheimer's disease, tear fluid contained elevated β -amyloid [15] and in glaucoma, the optic nerve head is enlarged [7]. These observations have become meaningful in the context of the ocular glymphatic clearance system. Hence, more focus should be given to in vivo human studies to validate and expand these findings.

3 Objective of the study

The objective of this study was to investigate the driving mechanisms and pulsatility of the human ocular glymphatic system by analysing optical flow in the human eye using fMREye data previously collected by the OFNI group. The analysis was performed in the VLF and RESP pulsality bands.

4 Materials and Methods

This study was conducted on fMREye imaging data sets acquired from 10 healthy control subjects (age 44.1 ± 13.4 years). The dataset had been collected earlier by the OFNI group and received approval from the Regional Ethics Committee of the Northern Ostrobothnia Hospital District (FIMEA/2021/002535). The work was conducted at Oulu University Hospital in partnership with the University of Oulu, Finland. All subjects gave written informed consent in line with the Declaration of Helsinki, and all study procedures complied with relevant ethical and institutional regulations. The sample included healthy volunteers as well as patient groups who met the eligibility criteria for multimodal imaging, which required no ongoing use of regular medications, being non-smokers, and not being pregnant.

Regions of interest (ROIs) were defined using high-resolution structural images, which were obtained using T1-weighted images. ROIs were manually drawn in FSLeves. In order to focus on both left and right eye regions, four ROIs were defined as retina, optic nerve, Optic nerve head and vitreous in each. Subsequently, voxel-wise flow fields were computed using a multi-resolution 3D Lucas–Kanade optical flow algorithm [6] was applied. The velocity magnitudes were computed as the Euclidean norm of the vector components. The sampling frequency of the MREG acquisition was 10 Hz, which was used in the temporal analyses of the signals.

All analyses were performed using the MATLAB(R2019) environment, and FSLeves was used for visualisation and qualitative assessment. As the MRI data were stored in NIfTI format (.nii.gz), the MATLAB NIfTI toolbox was used to import both the imaging and ROI mask files, ensuring compatibility and ease of analysis. For each ROI, temporal variation of velocity magnitude and direction were extracted and analysed separately for the VLF and RESP bands. Flow magnitude correlations between ROIs were quantified using the Pearson correlation coefficient, while directional synchrony was assessed using circular correlation analysis. Other statistical operations and visualisations were implemented using built-in MATLAB(R2019) functions. Additionally, Python scripts were used for automating certain operations (Eg: applying the optical flow algorithm, saving files, etc).

4.1 Data Aquisition

All MRI data were acquired on a Siemens MAGNETOM Vida 3T system (Siemens Healthineers, Erlangen, Germany) using a 64-channel head coil. Functional scans employed the Siemens ep2d gradient-recalled echo-planar imaging (GR-EPI) BOLD sequence

with the following settings: repetition time (TR) = 100 ms, echo time (TE) = 15 ms, echo train length = 32, flip angle = 15° , and an in-plane matrix size of 128×128 , providing an in-plane resolution of 2.64×2.64 mm. Each acquisition, each 3 mm thick, two axial slices, each 3 mm thick.

High-resolution anatomical images for registration and structural reference were obtained using T1-weighted MPRAGE (0.9-mm isotropic voxels; TR = 1900 ms; TE = 2.49 ms; flip angle = 9°) and T2-weighted SPC sequences (0.9-mm isotropic voxels; TR = 3200 ms; TE = 412 ms; flip angle = 90°), both acquired with a 240-mm field of view.

4.2 Preprocessing

Data processing and analysis were carried out using FSL (v5.0.9; BET; Jenkinson et al., 2012), AFNI (v2; Cox, 1996), and MATLAB (R2019). Functional images were preprocessed following the conventional FSL pipeline. The initial 20 volumes were removed to ensure signal stabilisation, and a high-pass temporal filter with a 0.008 Hz (125 s) cutoff was applied to eliminate slow drifts. Head motion was corrected using FSL’s MCFLIRT tool (v5.0.8; Jenkinson et al., 2002). To further mitigate brief artefactual fluctuations, AFNI’s 3dDespike was used to detect and attenuate signal spikes. Subsequent band-pass filtering was performed to extract the VLF (0.01–0.1 Hz) and RESP (0.2–0.4 Hz) frequency components. The cardiac band was not analysed in this work, consistent with previous findings from the OFNI group [9], which demonstrated that the eye regions exhibit only minimal cardiac-related pulsation compared to the brain regions and cerebrospinal fluid (CSF). In contrast, they have observed strong pulsations in the VLF and the RESP in the ocular regions. Consequently, the CARD component was considered less relevant for this optical flow analysis in the eye region, where VLF and RESP oscillations dominate.

4.3 Optical flow analysis

The optical flow output consisted of three directional velocity components (x , y , z). Because the dataset included only two slices in the z -axis, all directional analyses were restricted to the x – y plane.

Optical flow estimates motion by identifying spatial or temporal variations in image intensity or structural patterns between successive frames. In neuroimaging, this method is highly sensitive to subtle dynamic changes within tissues or fluid-filled spaces that cannot be captured using static MRI. When applied to fMREye acquisitions, optical flow calculates voxel-wise displacement from one frame to the next, producing a three-dimensional motion vector (dx , dy , dz) for each voxel at each time point t . These data collectively form a 5D motion field, defined by spatial position (x , y , z), vector components

(dx, dy, dz) , and temporal frame index t .

Optical flow methods have advanced considerably over time, with recent developments in machine learning leading to powerful deep learning-based approaches. Early classical techniques include the Horn–Schunck method (1981), which estimates dense flow under the assumption of smoothly varying motion, and the Lucas–Kanade method (1981), which computes local flow within small neighbourhoods and is well suited for sparse or semi-dense estimation. Subsequent contributions, such as the Farnebäck algorithm (2003), introduced polynomial neighbourhood modelling to generate dense flow with a favourable balance between speed and accuracy. More recently, deep learning has transformed the field: networks such as PWC-Net (2018) enable robust multi-scale motion estimation, while models like RAFT (Recurrent All-Pairs Field Transforms, 2020) provide highly accurate dense flow by combining global correlation with iterative refinement, allowing them to handle even large displacements.

In this study, we used the Lucas-Kanade method based on the OFNI groups Rajna et al.(2019) work [6].

5 Results

Optical flow analysis was performed on the VLF and RESP pulsation bands separately. To quantify the flow magnitude and the directional variation in the ocular glymphatic system, four important regions of interest (ROIs) were selected: retina, optic nerve(ON), optic nerve head (ONH) and vitreous for each eye. The results are presented separately for each band for ease of understanding.

5.1 Analysis on VLF band

Optical flow velocity magnitudes visualised a similar temporal variation pattern over the selected ROIs (figure 5.1), which indicates a comparable dynamic behaviour. However, quantitative comparison revealed a significant difference between the left and right ROIs. For example, the optic nerve ROI indicated very low inter-eye correlation ($r=0.16$), whereas the highest inter-eye correlation was observed in both vitreous and retina ROIs ($r=0.55$), as illustrated in the flow magnitude correlation matrix (figure 5.2). In contrast, the intra-eye velocity magnitude coupling between the optic nerve head and the optic nerve demonstrated moderate correlation (left eye: $r=0.5$; right eye: $r=0.67$). Similarly, the retina and optic nerve-head indicated considerable coupling (left eye: $r=0.61$; right eye: $r=0.65$). Further, the mean velocity magnitude and standard deviation(SD) are illustrated in the table 5.1. The optic nerve head exhibited the highest mean magnitude as well as the highest SD(L: 0.018 ± 0.015 ; R: 0.022 ± 0.017). In contrast, the optic nerve showed the lowest mean value and the SD (L: 0.006 ± 0.004 ; R: 0.012 ± 0.008). The other two ROIs showed moderate magnitude and SD.

ROI name	Mean	Standard deviation
L Vitreous	0.013	0.011
L Retina	0.014	0.012
L ONH	0.018	0.015
L ON	0.006	0.004
R Vitreous	0.015	0.012
R Retina	0.014	0.011
R ONH	0.022	0.017
R ON	0.012	0.008

Table 5.1: Mean and standard deviation of ROIs' velocity magnitude-VLF(vox/s)

Optical flow mean velocity direction (figure 5.3) exhibited a considerably complex temporal variation compared to the magnitude plot. Specifically, the left and right eyes indicated notable differences in the temporal signals across all ROIs. This is further

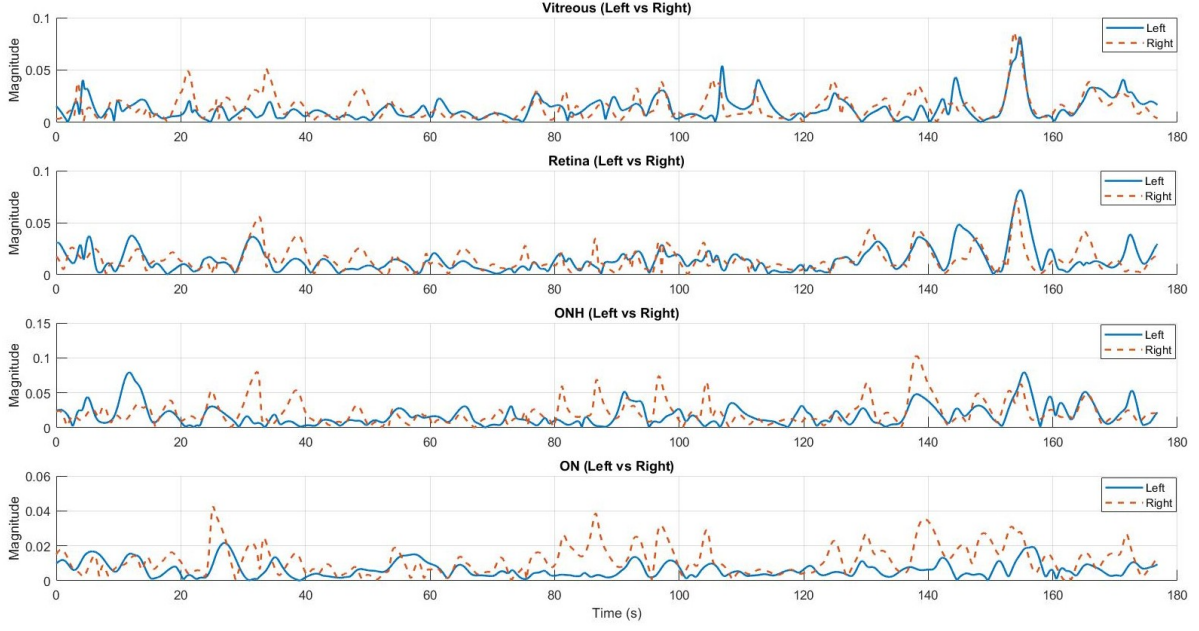


Figure 5.1: Variation of optical flow magnitude in left and right eyes-VLF (vox/s)

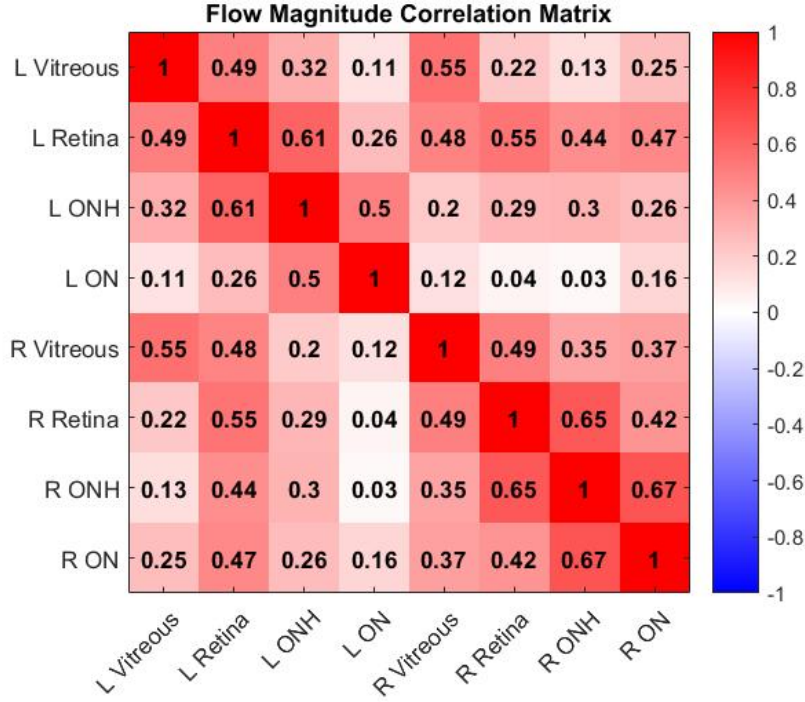


Figure 5.2: Magnitude correlation in left and right eyes-VLF

illustrated in the directional synchrony matrix (figure 5.4), where lower values were observed. For instance, the right vitreous showed negative synchrony with right optic nerve and right optic nerve head (-0.1 and -0.2), which was not observed in the left eye. Additionally, the Optic nerve head demonstrated moderate synchrony with the Optic nerve and the Retina. To examine the instantaneous switching of the flow direction, the

unwrapped angles were plotted for the optic nerve (figure 5.5) and the optic nerve head (figure 5.6). These plots revealed that, despite sudden changes in the original directional signals, the cumulative direction continues to increase, as shown in the window 01 of Figure 5.5. This is similar to a swirling flow. Additionally, actual sudden direction switchings were also highlighted in the unwrapped plot, as in window 02 of Figure 5.5.

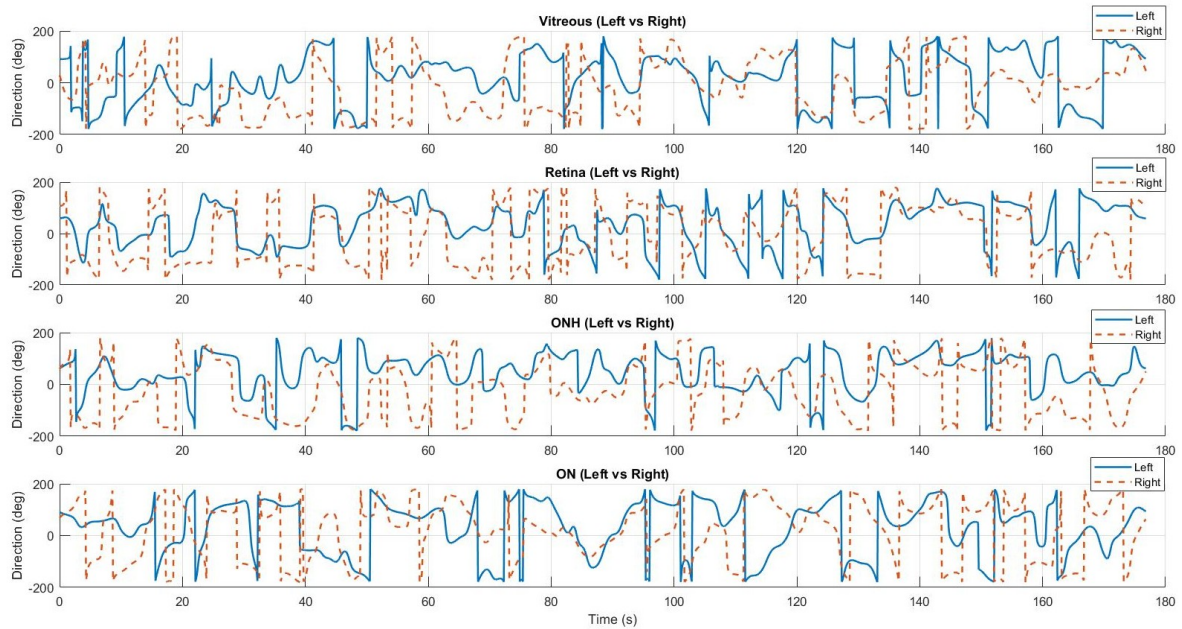


Figure 5.3: Variation of optical flow direction in left and right eyes-VLF

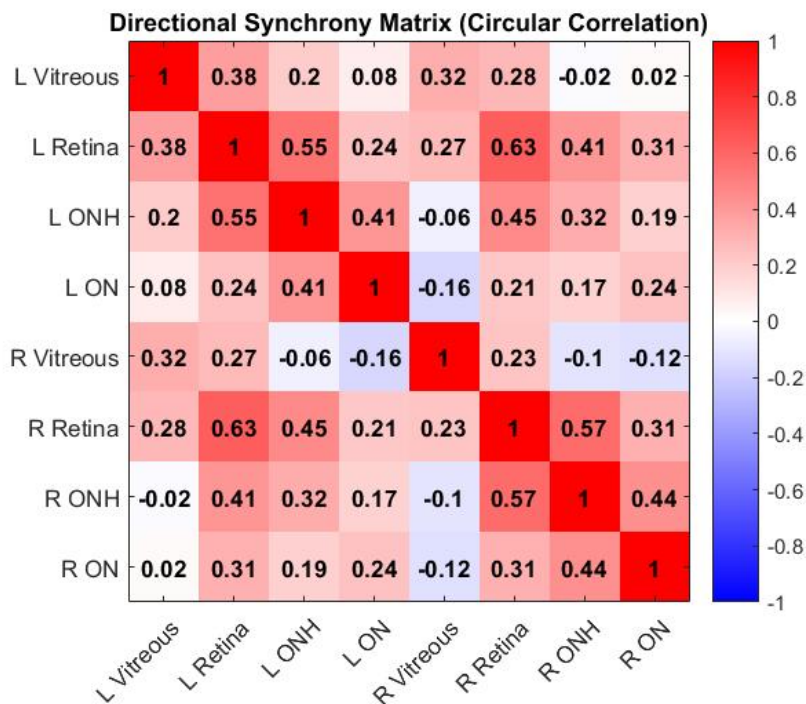


Figure 5.4: Directional synchrony of optical flow in left and right eyes- VLF

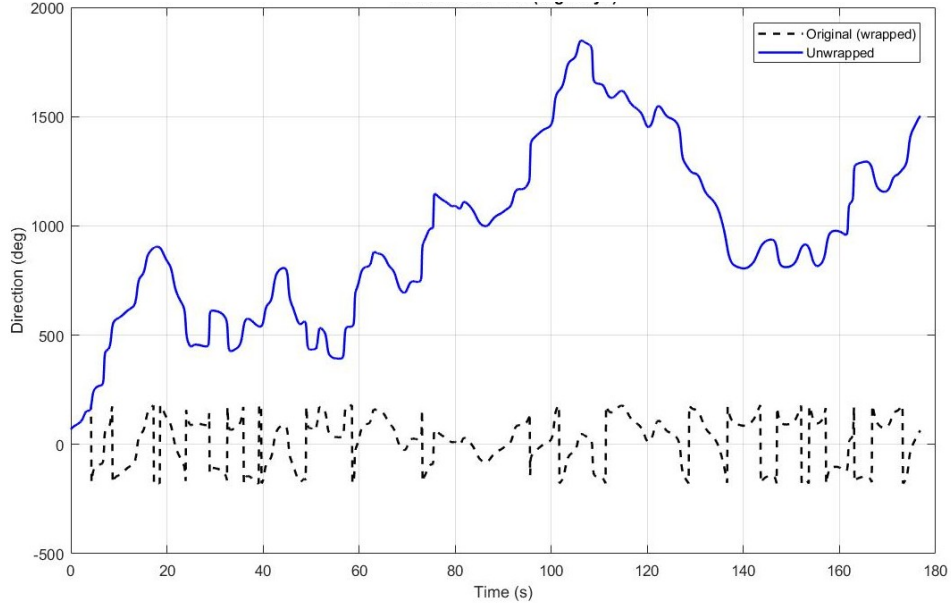


Figure 5.5: Unwrapped phase angle measured from the optic nerve-VLF

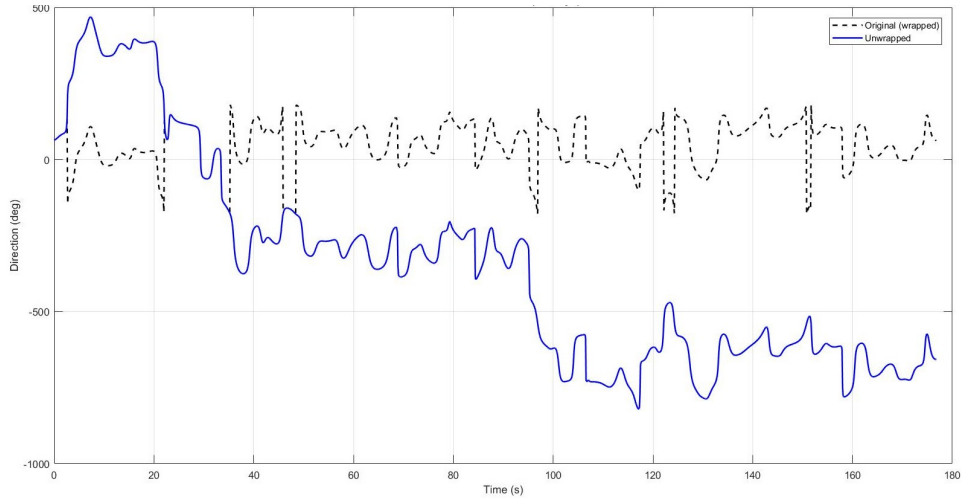


Figure 5.6: Unwrapped phase angle measured from the optic nerve head-VLF

Further, to investigate the plots quantitatively, the mean direction and its standard deviation were calculated as shown in the table 5.2. The mean directions of the left eye ROIs are quite similar, with positive mean angles, while the right eye showed mostly negative angles. Further, the standard deviation of both eyes is considerably high, where right optic nerve head has 154.03 degree is the highest.

5.2 Analysis on RESP band

The temporal variation in the magnitude of the optical flow velocity in the respiratory band also showed comparable patterns across the ROIs, as illustrated in the magnitude plot (figure 5.7). However, the left eye retina and optic nerve head exhibited higher

ROI name	Mean	Standard deviation
L Vitreous	62.97	109.40
L Retina	57.53	89.85
L ONH	65.95	70.27
L ON	95.85	95.62
R Vitreous	-81.68	135.47
R Retina	-165.91	105.82
R ONH	-55.09	154.03
R ON	108.90	101.60

Table 5.2: Mean and standard deviation of ROIs' velocity direction-VLF(vox/s)

amplitudes than those of the right eye. The correlation matrix (figure 5.8) revealed several important interocular correlations, particularly between right and left vitreous and optic nerve head ($r=0.51$). Additionally, intraocularly, the right optic nerve head showed strong correlations with the right retina ($r = 0.81$) and the right optic nerve ($r = -0.75$). A similar trend was also observed in the left optic nerve head, which was correlated with the left retina ($r = 0.72$). Interestingly, other ROIs demonstrated relatively less intraocular correlations within this band.

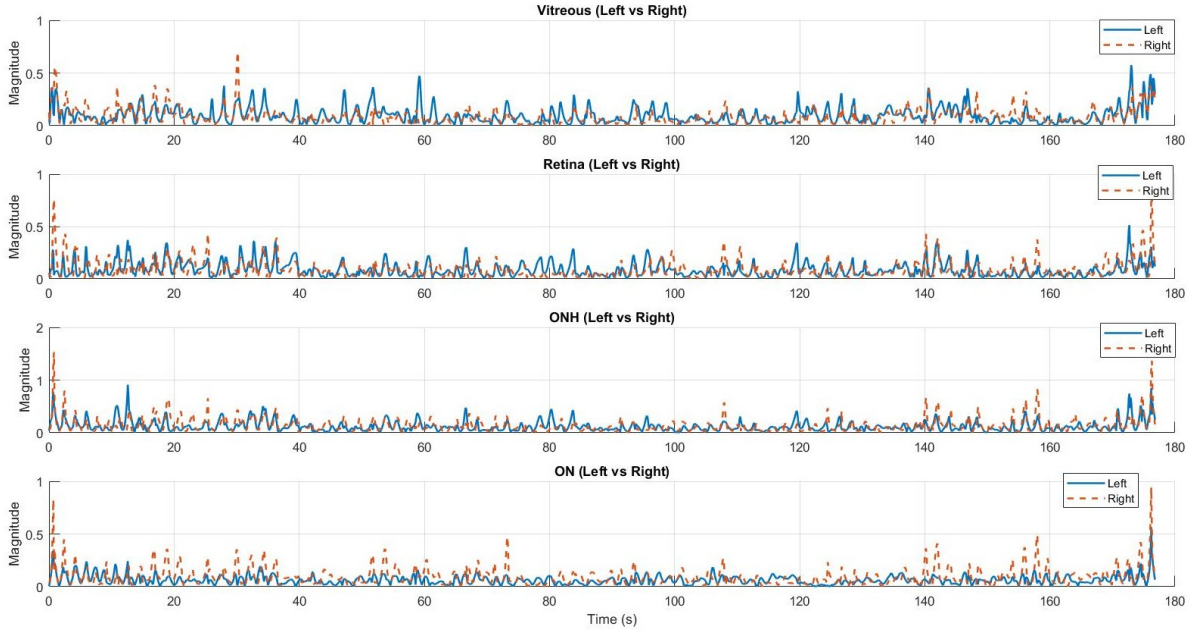


Figure 5.7: Variation of optical flow magnitude in left and right eyes-RESP (vox/s)

The mean and standard deviation of the optical flow velocity magnitude are presented in Table 5.3. The highest mean velocities were observed in the optic nerve head ($L = 0.127$, $R = 1.37$), which also showed the highest variability ($SD: L = 0.106$, $R = 0.131$). In contrast, the left optic nerve exhibited the lowest mean velocity (mean = 0.065, $SD = 0.043$).

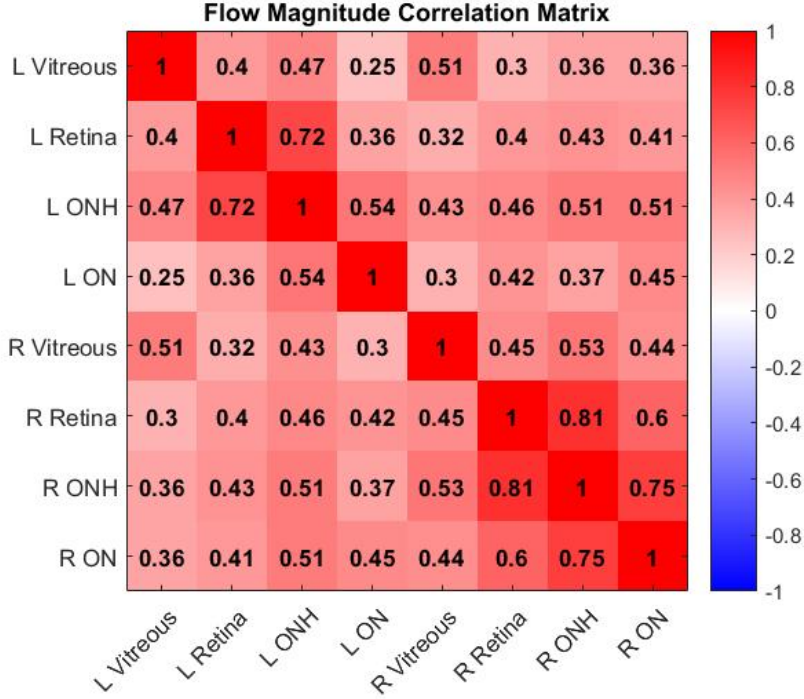


Figure 5.8: Magnitude correlation in left and right eyes-RESP

ROI name	Mean	Standard deviation
L Vitreous	0.098	0.078
L Retina	0.089	0.070
L ONH	0.127	0.106
L ON	0.065	0.043
R Vitreous	0.088	0.074
R Retina	0.090	0.081
R ONH	0.137	0.131
R ON	0.095	0.082

Table 5.3: Mean and standard deviation of ROIs' velocity magnitude-RESP band

The optical flow direction in the respiratory band exhibited a complex temporal pattern in the considered ROIs with notable dissimilarities between the left and right eyes (figure 5.9). This is further supported by the directional synchrony matrix (figure 5.10), which showed generally lower synchrony values. The only significant synchrony was observed between the retina and the optic nerve head (left: $r=0.6$, right: $r=0.75$). The mean direction of the velocity vector showed negative values and relatively high variability, as reflected by the large standard deviations (R retina = 110.04, L ON = 108.34).

The unwrapped direction plot of the optic nerve head (Figure 5.12) and optic nerve (Figure 5.11) demonstrated a cumulative angle increment despite the abrupt switching in the original wrapped plots. As illustrated in window 01 of Figure 5.12, the continuous

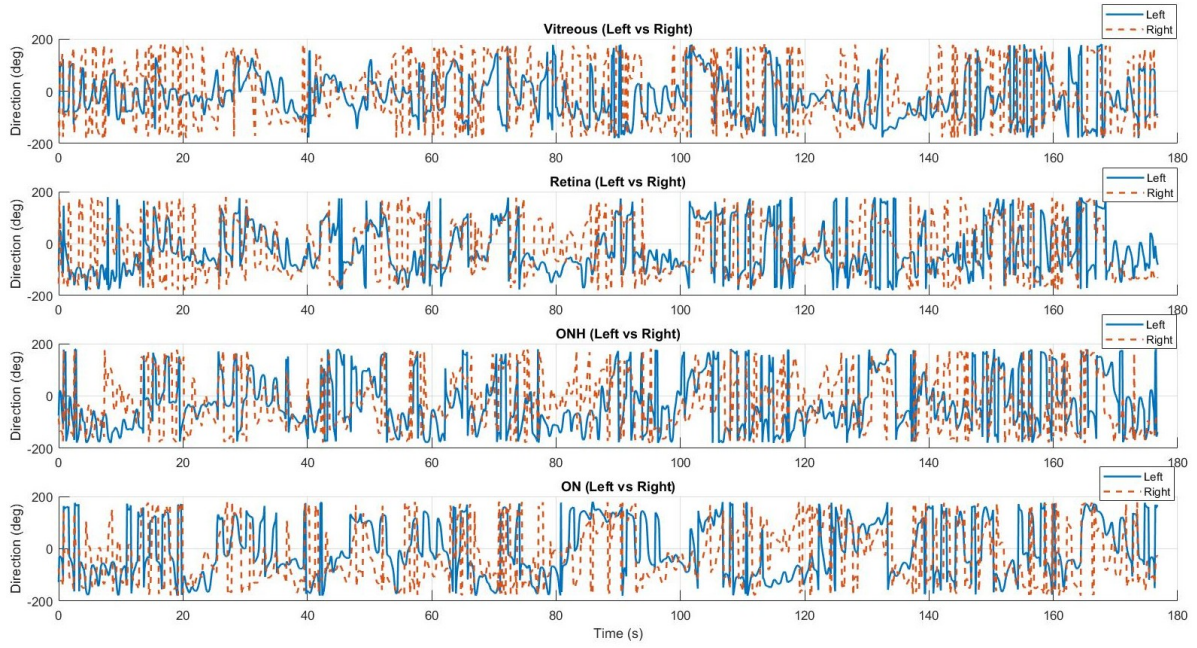


Figure 5.9: Variation of optical flow direction in left and right eyes-RESP

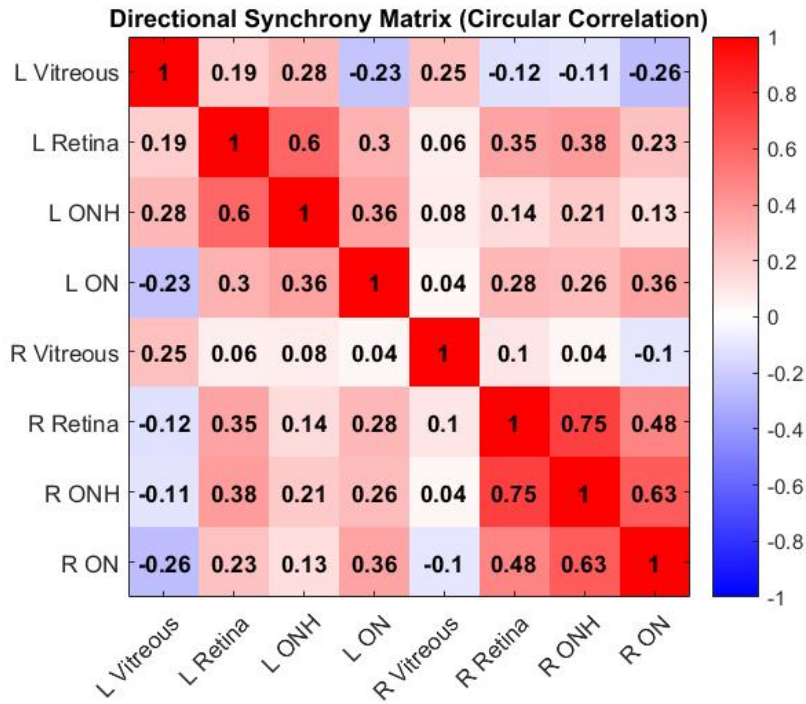


Figure 5.10: Directional synchrony of optical flow in left and right eyes- RESP

increase in the unwrapped angle indicated a swirl-like behaviour. In contrast, window 02 shows a true directional reversal occurring within approximately 3 seconds.

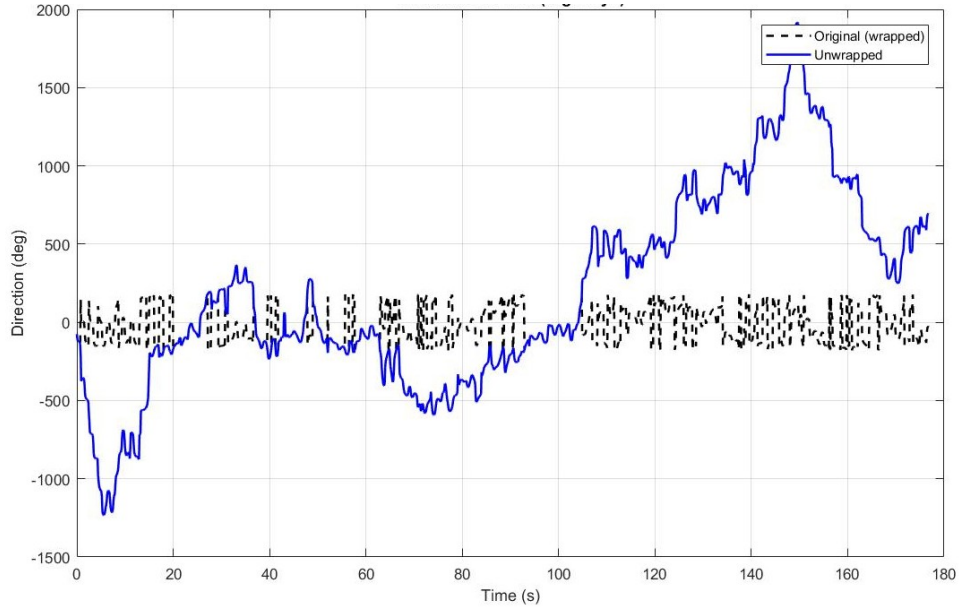


Figure 5.11: Unwrapped phase angle measured from the optic nerve-VLF

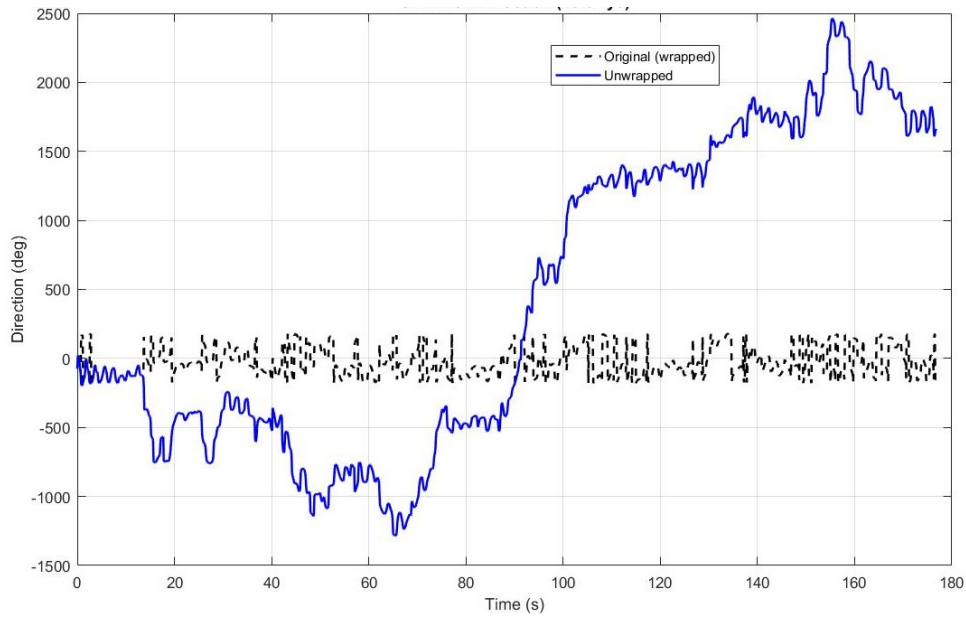


Figure 5.12: Unwrapped phase angle measured from the optic nerve head-VLF

5.3 Comparison of VLF and RESP bands in retina and optic nerve

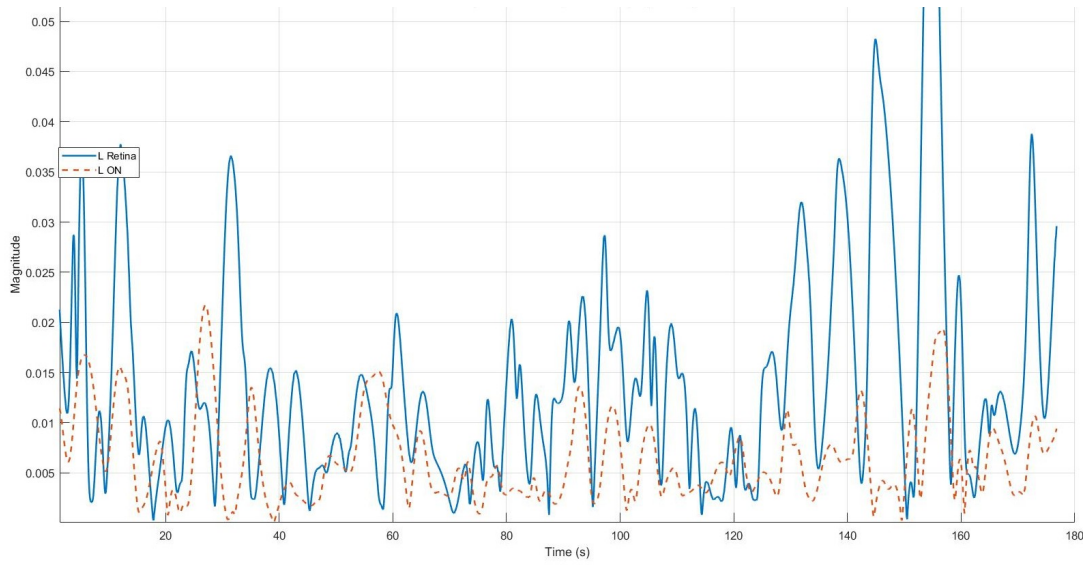
The VLF and RESP band signals were analysed in the optic nerve and retina, with the results presented separately for the left and right eyes.

The temporal variation of the optical flow magnitude in the VLF band for the left eye retina and optic nerve is shown in Figure 5.13(a), and for the right eye in Figure 5.13(b). The left retina demonstrated considerably higher magnitudes compared to the optic nerve; however, during certain time intervals, the optic nerve exhibited higher

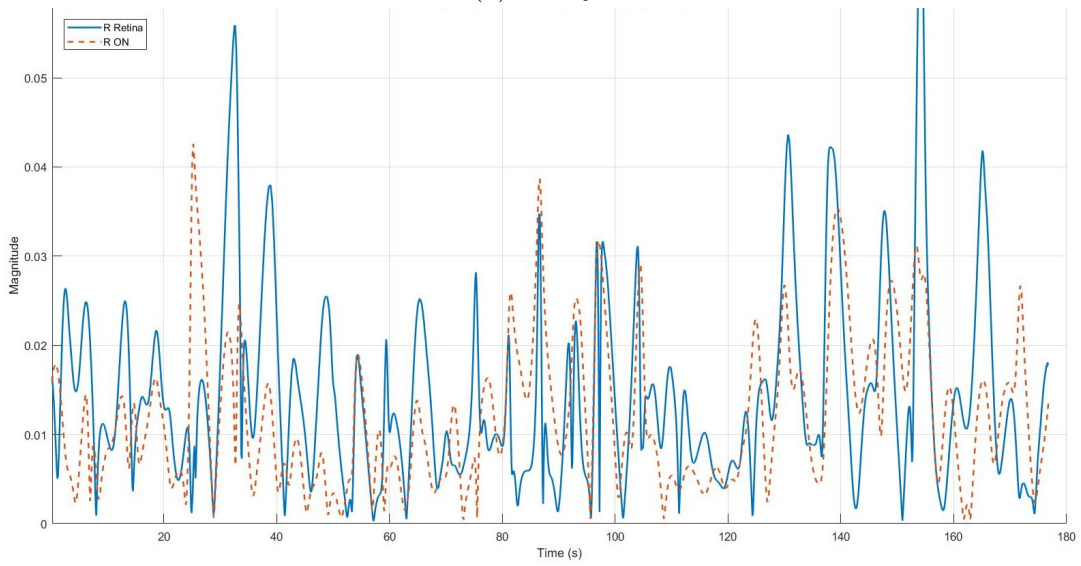
ROI name	Mean	Standard deviation
L Vitreous	-38.09	84.74
L Retina	-73.84	85.33
L ONH	-97.88	90.43
L ON	-116.77	108.34
R Vitreous	-169.11	95.30
R Retina	-152.97	110.04
R ONH	-126.48	105.09
R ON	-134.80	95.89

Table 5.4: Mean and standard deviation of ROIs' velocity direction-RESP band

values. Interestingly, both plots revealed an underlying envelope pattern in the magnitude variation. The right eye showed similar behaviour to the left, though with generally higher overall intensity. The optical flow directional behaviour in the same ROIs is illustrated in Figure 5.14(a) for the right eye and Figure 5.14(b) for the left eye. The temporal variations appeared complex, yet both sides indicated comparable envelope-like modulation patterns in the directional changes. Importantly, the analysis of the RESP band also demonstrated similar behaviour in these two ROIs for both eyes. Furthermore, the underlying envelopes were clearly visible in both the magnitude and directional plots (Figure 5.14), similar to the VLF band. However, the overall signal amplitudes in the VLF and RESP bands showed rather complex behaviour in the retina and optic nerve when comparing both eyes.

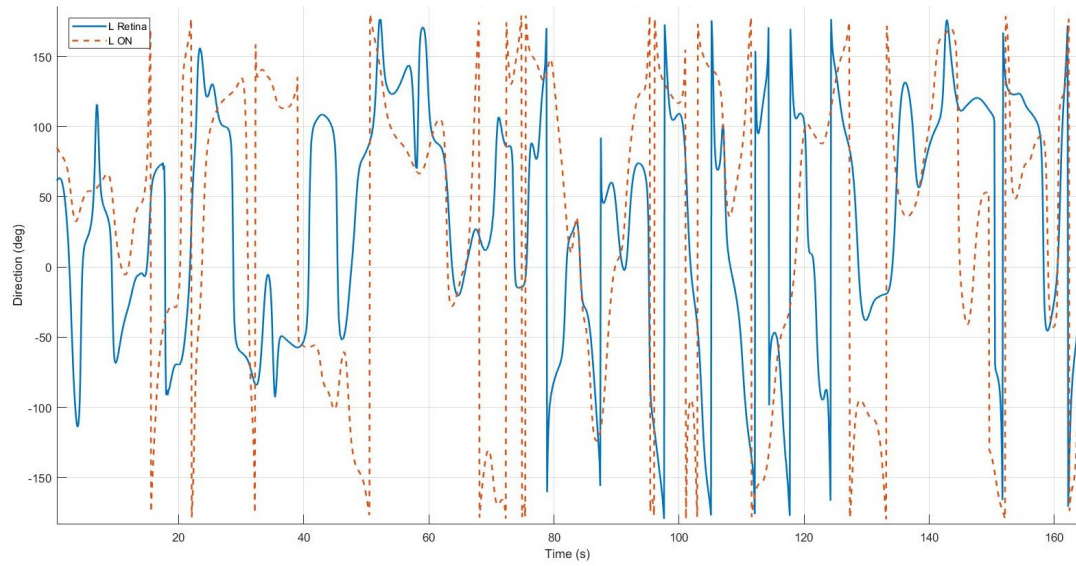


(a) left eye

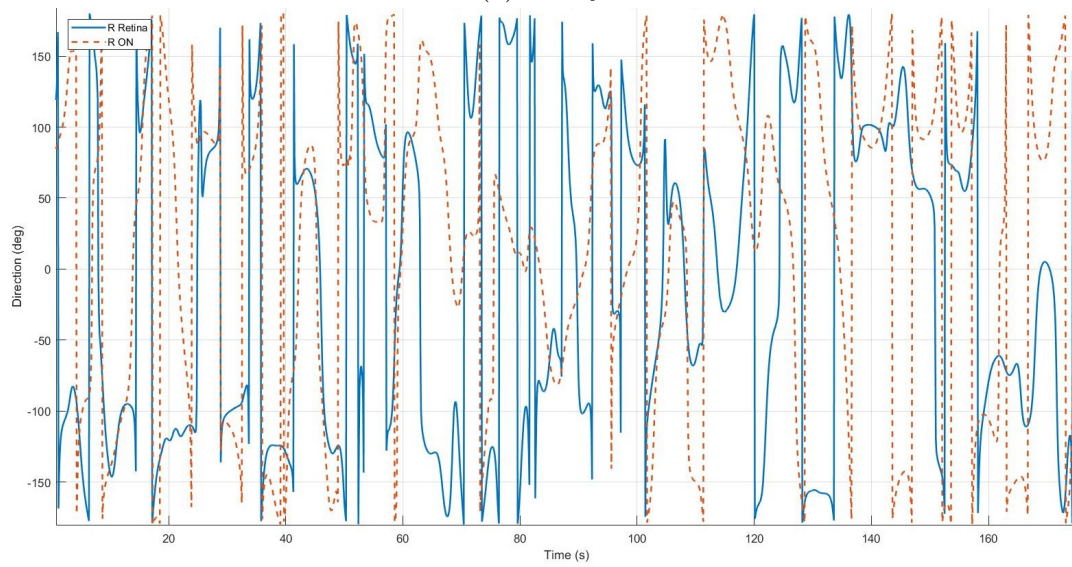


(b) right eye

Figure 5.13: Comparison of magnitude in retina and optic nerve-vlf



(a) left eye



(b) right eye

Figure 5.14: Comparison of direction in retina and optic nerve-vlf

6 Discussion

This study explored the characteristics of two major physiological pulsations, very low frequency and respiratory, in the human ocular glymphatic system using fMREye data. Four key regions of interest were examined: the optic nerve, optic nerve head, retina and vitreous. The cardiac band was omitted as it was found to be less prominent in the eye regions in previous studies. The analysis mainly focused on investigating the synchrony and inter-eye correlations of dynamics across the selected ROIs for the understanding of the underlying driving mechanisms of ocular glymphatic clearance.

The temporal variation of both magnitude and direction plots revealed a notable difference between the left and right eyes' dynamics. The plots demonstrated not only a lack of synchrony but also significant amplitude disparities between the two sides. This reduced inter-eye synchrony was observed in both the VLF and RESP bands, suggesting a degree of functional independence between the left and right ocular clearance. In contrast, the intraocular correlations, particularly between the optic nerve head, optic nerve and retina, showed substantial Synchrony, implying coordinated fluid dynamics along the introcular pathway along the optic nerve head. The optic nerve head exhibited the highest mean velocity magnitude and variability across both eyes and frequency bands, suggesting highly dynamic activity in this region. Given the optic nerve head's role as a critical junction between the introcular fluid flow and CSF flow through the optic nerve sheath from the cerebral glymphatic system. So, these observations may indicate bidirectional pressure or volume propagation across the laminar barrier, which is consistent with the previously proposed bidirectional ocular glymphatic exchange model. In contrast, the optic nerve exhibited lower magnitudes and variance, indicating a more stable flow region along the sheath. This is possibly reflecting adaptation to the cerebral fluid dynamics.

Furthermore, directional analysis of both bands revealed high variability in the ocular flow direction, as indicated by the higher standard deviations and wrapped angles. This suggests that the ocular fluid flow is not purely linear but oscillating dynamically, potentially driven by a cyclic deformation of ocular tissues (due to pulsatility). The unwrapped phase-angle plots showed gradual cumulative changes in the direction over considerable time intervals, consistent with swirl-like (slow rotation) fluid motion. Further, several periodical but fast (less than 3 seconds) reversals in direction indicate potential bidirectional flow within the optic nerve region. These findings support the concept of a complex, cyclic/ vortex-like or oscillatory motion within the ocular fluid transportation system. However, future work should validate this interpretation by analysing both vector direction and magnitude simultaneously in a larger cohort of healthy subjects. In addition to that, the presence of an envelope-like modulation pattern superimposed on

the magnitude and directional plots of both frequency bands was observed. This slower oscillation occurred at frequencies below the VLF band but needs further investigation for further precision.

Overall, the VLF oscillations appeared smoother and more globally synchronised, likely reflecting vasomotor-driven flow, whereas the RESP oscillations were sharper and more transient, consistent with respiratory-induced intracranial pressure fluctuations. The observed domination of VLF and RESP components within the ocular structures aligns with the current understanding of the glymphatic-like clearance mechanism extending from the brain into the optic pathway. The highly dynamic activity and swirl-like motion observed in the optic nerve head further support the hypothesis of bidirectional flow and CSF-ISF exchange through the laminar barrier.

Nevertheless, this study has several limitations. The analysis was conducted on a 10-subject dataset with only two slices, limiting spatial resolution in the z-direction. Although consistent patterns were observed across both frequency bands, validation with multiple healthy subjects is essential to confirm these findings. Furthermore, the optical flow vectors represent apparent motion derived from intensity variations in images, rather than direct fluid motion, which introduces interpretational constraints. Future work should incorporate multi-slice, multi-subject acquisition combined with simultaneous physiological monitoring to strengthen the physiological interpretation of the optical flow dynamics. Further, analysis of cerebral ROIs should be incorporated by including on CARD band in the cerebral regions.

Despite these limitations, the present results demonstrate that quantitative optical flow analysis of fMREye data provides a promising and powerful non-invasive approach to study physiological pulsations and understand the ocular glymphatic clearance system.

7 Conclusions

This study investigated the characteristics of very low frequency and respiratory physiological pulsations within the ocular glymphatic system using MREG and EPI data by applying optical flow analysis. Analysis was performed by selecting key ocular regions: vitreous, retina, optic nerve head, and optic nerve. The study successfully demonstrated that ocular tissues exhibit distinct, measurable flow dynamics corresponding to these physiological oscillations.

The results revealed that ocular optical flow is not uniform across some regions. Specially, the left and right eyes showed asynchronous and amplitude-varying behaviour, suggesting a degree of functional independence between the two sides of the ocular glymphatic systems. However, within the eye, strong correlations between the optic nerve head, optic nerve, and retina indicated synchronised intraocular fluid dynamics along the optic pathway. The optic nerve head emerged as a particularly dynamic region, exhibiting the highest magnitudes and variances, suggesting active bidirectional flow exchange at the interface between the ocular and cerebral compartments.

Directional analyses revealed the possibility of a complex fluid motion, including swirl-like and bidirectional patterns, especially in the optic nerve and optic nerve head. These patterns support the concept of interstitial cerebrospinal fluid exchange. Additionally, the presence of slower envelope-like modulations in both VLF and RESP bands suggests underlying coupling in ocular pulsation dynamics.

Overall, these findings reinforce the emerging concept that the ocular glymphatic bidirectional pathway is an integral extension of the brain’s waste-clearance system, modulated by both vasomotor and respiratory pulsations. The optical flow framework presented here demonstrates a non-invasive, quantitative approach to assessing these physiological processes *in vivo*.

Based on the results, several recommendations can be made for future research. Mainly, validation across multiple healthy subjects and patient groups (e.g., glaucoma or Alzheimer’s disease) is essential to confirm the findings and potential diagnostic value of these observations. Further, data acquisitions should employ multi-slice sequences to capture full volumetric ocular and optic nerve dynamics, reducing plane motion artefacts, and simultaneous cardiac and respiratory recordings would allow more accurate phase-aligned interpretation of pulsatile flow components. Finally, the observed asynchrony and directional variability could be explored as potential biomarkers for impaired glymphatic clearance in so should be included patient groups.

Bibliography

- [1] C. Delle, X. Wang, and M. Nedergaard, “The ocular glymphatic system—current understanding and future perspectives,” *International Journal of Molecular Sciences*, vol. 25, no. 11, p. 5734, 2024.
- [2] J. J. Iliff, M. Wang, Y. Liao, B. A. Plogg, W. Peng, G. A. Gundersen, H. Benveniste, G. E. Vates, R. Deane, S. A. Goldman, E. A. Nagelhus, and M. Nedergaard, “A paravascular pathway facilitates csf flow through the brain parenchyma and the clearance of interstitial solutes, including amyloid β ,” *Science Translational Medicine*, vol. 4, p. 147ra111, Aug 2012.
- [3] X. Wang, N. Lou, A. Eberhardt, Y. Yang, P. Kusk, Q. Xu, B. Förster, S. Peng, M. Shi, A. Ladrón-de Guevara, C. Delle, B. Sigurdsson, A. L. R. Xavier, A. Ertürk, R. T. Libby, L. Chen, A. S. Thrane, and M. Nedergaard, “An ocular glymphatic clearance system removes β -amyloid from the rodent eye,” *Science Translational Medicine*, vol. 12, no. 536, p. eaaw3210, 2020.
- [4] Y. Xu, L. Cheng, L. Yuan, Q. Yi, L. Xiao, and H. Chen, “Progress on brain and ocular lymphatic system,” *Biomedical Research International*, vol. 2022, p. 6413553, Nov 15 2022.
- [5] V. Kiviniemi, X. Wang, V. Korhonen, T. Keinänen, T. Tuovinen, J. Autio, P. LeVan, S. Keilholz, Y.-F. Zang, J. Hennig, and M. Nedergaard, “Ultra-fast magnetic resonance encephalography of physiological brain activity – glymphatic pulsation mechanisms?,” *Journal of Cerebral Blood Flow & Metabolism*, vol. 36, pp. 1033–1045, Jun 2016.
- [6] Z. Rajna, L. Raitamaa, T. Tuovinen, J. Heikkilä, V. Kiviniemi, and T. Seppänen, “3d multi-resolution optical flow analysis of cardiovascular pulse propagation in human brain,” *IEEE Transactions on Medical Imaging*, vol. 38, no. 9, pp. 2028–2036, 2019.
- [7] X. Wang, C. Delle, W. Peng, V. Plá, M. Giannetto, P. Kusk, B. Sigurdsson, S. Sakurai, A. Sweeney, Q. Sun, T. Du, R. T. Libby, and M. Nedergaard, “Age- and glaucoma-induced changes to the ocular glymphatic system,” *Neurobiology of Disease*, vol. 188, p. 106322, 2023.
- [8] A. Bakdalieh, L. M. Khawaja, and M. Yu, “Dark adaptometry as a diagnostic tool in retinal diseases: Mechanisms and clinical utility,” *Journal of Clinical Medicine*, vol. 14, no. 11, p. 3742, 2025.

- [9] S. Ebrahimi, J. Tuunanen, V. Saarela, M. Honkamo, N. Huotari, L. Raitamaa, V. Korhonen, H. Helakari, M. Järvelä, M. Kaakinen, L. Eklund, and V. Kiviniemi, “Synchronous functional magnetic resonance eye imaging, video ophthalmoscopy, and eye surface imaging reveal the human brain and eye pulsation mechanisms,” *Scientific Reports*, vol. 14, no. 1, p. 2250, 2024.
- [10] A. K. Denniston and P. A. Keane, “Paravascular pathways in the eye: Is there an ‘ocular glymphatic system’?,” *Investigative Ophthalmology & Visual Science*, vol. 56, no. 6, pp. 3955–3955, 2015.
- [11] X. Wang, H. Jiang, L. Li, L. Wang, L. Dong, and Y. Wang, “Bidirectional transport of cerebrospinal fluid along the optic nerve in mice: In vivo imaging and tracer studies,” *Journal of Magnetic Resonance Imaging*, vol. 57, no. 2, pp. 513–523, 2023.
- [12] H. E. Killer, H. R. Laeng, and P. Groscurth, “Lymphatic capillaries in the meninges of the human optic nerve,” *Journal of Neuro-Ophthalmology*, vol. 19, pp. 222–228, 1999.
- [13] Y. H. Yücel, M. G. Johnston, T. Ly, M. Patel, B. Drake, L. Gregory, M. Boulton, C. Hilberg, T. Wong, and N. Gupta, “Human ocular lymphatics and their distribution in the eye,” *Investigative Ophthalmology & Visual Science*, vol. 50, no. 10, pp. 4283–4291, 2009.
- [14] X. Yin, S. Zhang, J. H. Lee, H. Dong, G. Mourgos, G. Terwilliger, A. Kraus, L. H. Geraldo, M. Poulet, S. Fischer, T. Zhou, F. Shalima, M. F. S., T. Zhou, Y. Wang, S. Malloy, N. Rohner, L. Sharma, I. Salinas, C. Eichmann, A. Thomas, J.-L. Thomas, W. M. Saltzman, A. Huttner, C. Zeiss, A. Ring, A. Iwasaki, and E. Song, “Compartmentalized ocular lymphatic system mediates eye–brain immunity,” *Nature*, vol. 628, no. 8006, pp. 204–211, 2024.
- [15] M. M. van der Thiel, N. van de Sande, A. Meeusen, G. S. Drenthen, A. A. Postma, R. M. M. A. Nuijts, N. van der Knaap, I. H. G. B. Ramakers, C. A. B. Webers, W. H. Backes, M. Gijs, and J. F. A. Jansen, “Linking human cerebral and ocular waste clearance: Insights from tear fluid and ultra-high field mri,” *Neurobiology of Disease*, vol. 203, p. 106730, 2024.
- [16] S. R. Levendovszky, J. Flores, E. R. Peskind, L. Václavů, M. J. P. van Osch, and J. Iliff, “Preliminary investigations into human neurofluid transport using multiple novel non-contrast mri methods,” *Journal of Cerebral Blood Flow & Metabolism*, vol. 44, no. 12, p. 1580-1592, 2024.
- [17] R. W. F. Damasceno, J. A. P. Barbosa, L. R. C. Cortez, and R. B. Jr., “Orbital lymphatic vessels: immunohistochemical detection in the lacrimal gland, optic nerve,

fat tissue, and extrinsic oculomotor muscles,” *Arquivos Brasileiros de Oftalmologia*, vol. 84, no. 3, pp. 209–213, 2021.